THE CURVES STACK SATISFIES (RS*) ON PUSHOUTS

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Lemma more-morphisms-lemma-pushout-along-thickening (canonical tag 07RT). cited at proof lines 4910 and 4942. The pushout square; the fibres over the thickening agree.
- Equation equation-curves-over-proper-spaces (canonical tag 0D50). cited at proof line 4923. The embedding of the curves stack.
- Lemma lemma-spaces-RS-star (canonical tag 0D1J). cited at proof line 4926. The spaces stack (RS*) fact.
- Lemma spaces-morphisms-lemma-dimension-fibre-after-base-change (canonical tag 04NS). cited at proof line 4936. Relative dimension <= 1 preserved under base change.

Source reference. Original lemma 07RT: sources/stacks-chow/tags.txt, source lines 9878–9878. Cited at proof lines 4910 and 4942: the pushout square and the fibre agreement.

Source reference. Original equation 0D50: sources/stacks-chow/tags.txt, source lines 16470–16470. Cited at proof line 4923: the embedding.

Source reference. Original lemma 0D1J: sources/stacks-chow/tags.txt, source lines 16349–16349. Cited at proof line 4926: the spaces stack fact.

Source reference. Original lemma 04NS: sources/stacks-chow/tags.txt, source lines 6090–6090. Cited at proof line 4936: relative dimension preserved.

Lemma 0.2. *Let*

$$\begin{array}{ccc} T & \longrightarrow & T' \\ \downarrow & & \downarrow \\ S & \longrightarrow & S' \end{array}$$

be a pushout in the category of schemes where $T \to T'$ is a thickening and $T \to S$ is affine, see More on Morphisms, Lemma 07RT. Then the functor on fibre categories

$$\mathcal{C}urves_{S'} \longrightarrow \mathcal{C}urves_S \times_{\mathcal{C}urves_T} \mathcal{C}urves_{T'}$$

is an equivalence.

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Proof. Using the embedding (0D50), the description of the image, and the corresponding fact for $\mathcal{S}paces'_{fp,flat,proper}$ (Lemma 0D1J) this reduces to the following statement: given a morphism $X' \to S'$ of an algebraic space to S' which is of finite presentation, flat, proper then $X' \to S'$ has relative dimension ≤ 1 if and only if $S \times_{S'} X' \to S$ and $T' \times_{S'} X' \to T'$ have relative dimension ≤ 1. One implication follows from the fact that having relative dimension ≤ 1 is preserved under base change (Morphisms of Spaces, Lemma 04NS). The other follows from the fact that having relative dimension ≤ 1 is checked on the fibres and that the fibres of $X' \to S'$ (over points of the scheme S') are the same as the fibres of $S \times_{S'} X' \to S$ since $S \to S'$ is a thickening by More on Morphisms, Lemma 07RT. $\square$